Movie Memory Experiment

Report 1: Descriptive and Inferential Statistics

The Standard Deviants

1. Introduction

Our study investigated whether disrupting natural event boundaries during video viewing effect visual recognition memory. Participants watched YouTube Shorts clips under two conditions: **Natural Boundary (NB)**, where clips ended at a natural scene transition, and **Abrupt Boundary (AB)**, where clips were cut 1–5 seconds before the boundary. After viewing, participants completed a two-alternative forced-choice (2AFC) recognition test, identifying target frames (seen) from lure frames (unseen). Targets were drawn from either **before the boundary (BB)** or the **event middle (EM)** frames, allowing us to test whether any memory impairment is specific to boundary-adjacent frames or due to whole event.

2. Background

Human memory does not encode all moments equally. According to **Event Segmentation Theory** (EST; Zacks et al., 2007), the brain continuously builds a working model of the current situation and generates predictions about what will happen next. When incoming information violates these predictions a **prediction error** is generated, signalling the start of a new event. This triggers a model update that strengthens memory encoding for boundary-adjacent content, while making prior event information temporarily less accessible (Radvansky & Zacks, 2017). The result is the **boundary advantage**: recognition memory is better for frames near event boundaries than for frames from the event middle (Swallow, Zacks, & Abrams, 2009). By artificially cutting a clip before its natural boundary, the AB condition prevents the normal model-update process from completing, which should weaken the boundary-driven encoding boost and reduce memory for the disrupted content.

3. Dataset Description

A total of **185 participants** from IIIT Hyderabad community and are randomly assigned to one of two between-subjects conditions. In the **Natural Boundary (NB)** condition, participants watched video clips that ended at a natural event boundary. In the **Abrupt Boundary (AB)** condition, clips were cut 1–5 seconds before the natural boundary.

Each participant watched 45 video clips during encoding 40 unique YouTube Shorts and 5 repeated clips serving as **vigilance checks** (spacebar press on recognition). After encoding, participants completed a 2AFC recognition test for each unique clip, choosing between a **target** (seen) and a **lure** frame (unseen). Targets were either **BB** or **EM**. After each decision, participants rated confidence on a 1–5 scale. The measured variables are in Table 1.

Table 1. Measured Variables

Variable	**Type**	**Description**
Accuracy	Binary	Correctly identified target (1) or incorrect (0)
RT	Cont. (s)	Time from stimulus display to key press
Confidence	Ordinal	Self-rated certainty (1 = low, 5 = high)
Frame Type	Categorical	BB (near boundary) or EM (event middle)
Condition	Categorical	NB (natural boundary) or AB (abrupt boundary)
Enc. Time	Cont. (min)	Total time spent watching videos during encoding

4. Methods

4.1 Participant Demographics

A total of **185 participants** were recorded, of whom **171** had usable behavioural data files. Of these, **163** had valid demographic records; the remaining **22** were missing demographic information and are included in all core behavioural analyses but excluded from demographic-specific tests (H6). Both groups were well-matched on all demographic variables; a breakdown is given in Table 2.

Table 2. Participant Demographics by Condition

Variable	Category	AB	NB
Age	Range (years)	19–34	
	Mean (SD)	22.3 (2.3)	22.3 (2.4)
Gender	Male	55	58
	Female	27	23
Handedness	Right-handed	79	75
	Left-handed	3	6
Vision	Normal	44	40
	Corrected	38	39
	Uncorrected	0	2

4.2 Data Cleaning

Incomplete data. The file for participant sub42 (NB) contained only demographic output with no recognition trial data. Participant sub20 data was unavailable. Both were excluded from all analyses.

Excessive encoding time. The five repeated clips account for 5.6 minutes of total encoding time; assuming 25 seconds per repeated-clip recognition, participants spending more than 27.05 minutes in total were deemed inattentive. Four participants exceeded this threshold and were excluded (Table 3).

Table 3. Participants Excluded for Excessive Encoding Time

Participant	Condition	Enc. Time (min)
sub32	AB	27.33
sub36	AB	27.77
sub151	NB	27.16
sub161	NB	28.00

Response time outliers. Trials with RT < 0.2 s or RT > 60 s were excluded. One outlier was identified: sub28 (AB, Movie 8, RT = 71.09 s).

Response time cleaning. In some of the analysis we used the distributions and applied data cleaning. Trials with response times (RT) greater than $Mean + 4 \times SD$ were removed to minimize the impact of inattentiveness. This resulted in the removal of 54 trials (0.8% of total data).

Condition re-classification. Participant sub157 was labelled AB in the filename, but the video path column confirmed NB stimuli were viewed. Sub157 was re-classified to NB.

Missing demographic data. Twenty-two participants who did not fill the demographic form are retained in all core analyses but excluded from demographic balance tests (H6).

After all exclusions, the **final analysed sample comprised 166 participants** (AB: $n = 78$; NB: $n = 88$).

4.3 Hypotheses

Based on Event Segmentation Theory, we came up with six hypotheses.

H1 — Effect of Condition

IV	Viewing condition (NB vs. AB) — between subjects
DV	Mean recognition accuracy, mean RT (s)
H_0	No difference in accuracy or RT between NB and AB
H_A	Participants in the NB condition will show higher overall recognition accuracy and faster response times than participants in the AB condition.
Direction	One-tailed (NB > AB accuracy; NB < AB RT)
Test	Mann-Whitney U test & trial-level Regression (for RT)

H2 — Effect Larger for BB Frames

IV	Viewing condition (NB vs. AB) — between subjects
DV	Mean recognition accuracy for BB frames
H_0	No difference in BB accuracy between conditions
H_A	Recognition accuracy will be greater for before-boundary (BB) frames than for event-middle (EM) frames.Reflecting NB advantage
Direction	One-tailed (Accuracy : BB > EM)
Test	Independent-samples t-test

H3 — No Condition Effect for EM Frames

IV	Viewing condition (NB vs. AB) — between subjects
DV	Mean recognition accuracy for EM frames
H_0	Recognition accuracy for EM frames will not differ significantly between the NB and AB groups.
H_A	Significant difference in EM accuracy
Direction	Two-tailed (EM frames are away from the manipulation point)
Test	Independent-samples t-test

H4 — Confidence Calibration

IV	Trial outcome (correct / incorrect) — within subjects
DV	Mean confidence rating (1–5 scale)
H_0	Confidence equal for correct and incorrect trials within each condition (AB and NB)
H_A	Within each condition (AB and NB), confidence ratings will be higher for correctly recognised trials than for incorrectly recognised trials
Direction	One-tailed
Test	Shapiro-Wilk (normality) → Paired t-test (if normal) or Wilcoxon Signed-Rank (if non-normal), applied separately per condition

H5 — Confidence by Frame Type

IV	Frame type (BB vs. EM) — within subjects
DV	Mean confidence rating (1–5 scale)
H_0	Confidence equal for BB and EM trials
H_A	Confidence ratings will be higher for BB trials than for EM trials.
Direction	One-tailed
Test	Paired-samples t-test

H6 — Demographic Balance	
IV	Viewing condition (NB vs. AB) — between subjects
DV	Age; gender; handedness; vision status
H_0	The NB and AB groups will not differ significantly on age, gender distribution, handedness distribution, or vision status.
H_A	Demographic distributions differ significantly
Direction	Two-tailed
Test	Independent t-test (age); χ^2 test (categorical variables)

4.4 Statistical Analysis

Normality of distributions was verified using the Shapiro-Wilk test before each inferential test. Because the data violated normality for H1 (accuracy and RT), H2 (BB frames), and H3 (EM frames) across both groups, we used the non-parametric **Mann-Whitney U test** in all these cases. For Hypothesis 1, the participant-level comparison was supplemented by a trial-level OLS Linear Regression for RT to evaluate processing speed in greater detail. For H4, where the NB group's difference scores were non-normal ($W = 0.967, p = .025$), the Wilcoxon Signed-Rank test was substituted for the paired t-test. For H6 age, the Mann-Whitney U test was similarly applied. In addition to these participant-level tests, we fitted **trial-level Regression models** (OLS for RT and H5) to evaluate predictive effects without losing trial-level variability. The significance threshold was $\alpha = 0.05$ throughout.

5. Results

5.1 H1 — Main Effect of Condition

5.1.1 Accuracy

Prediction: We expected that participants in the NB condition would have better recognition accuracy than those in the AB condition.

Descriptive Statistics: Participants in the NB group identified targets more accurately (average = 87.4%, SD = 7.0%, median = 87.5%) than participants in the AB group (average = 84.0%, SD = 8.0%, median = 85.0%) showing that NB group performed better overall.

Assumption Testing: Shapiro-Wilk test confirmed that accuracy data for both groups was not normally distributed (AB: $W = 0.942, p < .001$; NB: $W = 0.914, p < .001$). So we used non-parametric test (Mann-Whitney U test) to compare the groups.

Inferential Statistics: The Mann-Whitney U test showed that the difference in accuracy between the two groups is statistically significant ($U = 2550.5, p = .0042$).

Conclusion: We reject the null hypothesis ($p < .05$) and accept the alternative hypothesis. Supporting the idea from Event Segmentation Theory that keeping natural event boundaries helps improve visual recognition memory.

5.1.2 Response Time

Prediction: We expected that participants in the NB condition would respond faster than those in the AB condition.

Data Cleaning: To make sure the data is accurate, we removed trials where the response time was very high (greater than $M + 4 \times SD$) as mentioned in 4.2.

Descriptive Statistics: Contrary to our prediction the NB group ($M = 5.75$ s, $SD = 1.61$ s, $Mdn = 5.55$ s) was not faster than the AB group ($M = 5.59$ s, $SD = 1.32$ s, $Mdn = 5.36$ s). But in fact the NB participants were slightly slower on average.

Assumption Testing: The Shapiro-Wilk test showed that the response time data was not normally distributed for both groups (AB: $W = 0.942, p = .0015$; NB: $W = 0.965, p = .0183$). So we used the non-parametric Mann-Whitney U test to compare the groups.

Inferential Statistics: The Mann-Whitney U test showed that the difference in response time between the two groups is not statistically significant ($U = 3246.0, p = .548$). We also used a trial-level OLS Linear Regression on all responses which showed that the condition was not a significant predictor of response time ($B = 0.162, p = .081$).

Conclusion: We fail to reject the null hypothesis ($p > .05$). This is important as it shows there is no "speed-accuracy trade-off." Meaning the NB group's higher accuracy was not because they took more time but because their memory was actually better.

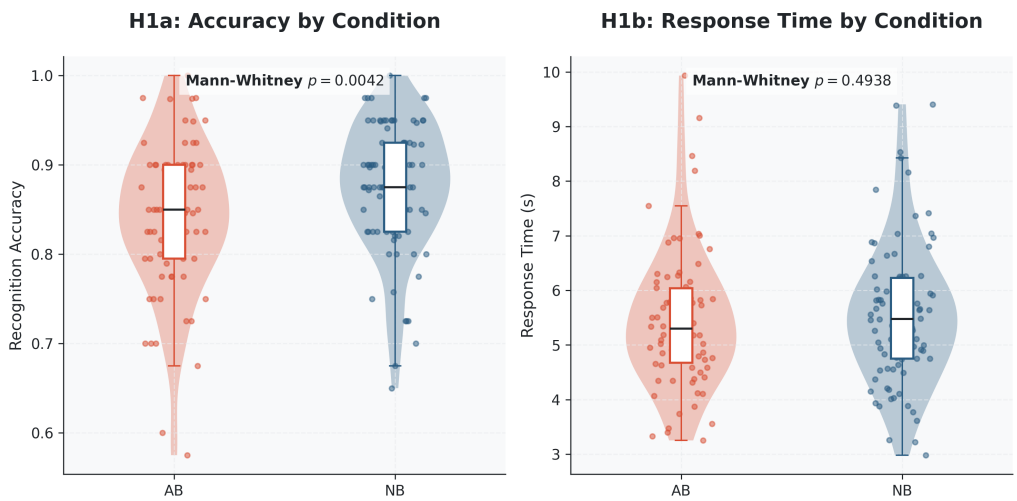

Figure 5.1 — H1: Mean recognition accuracy and mean response time by condition (NB vs. AB).

5.2 H2 — Recognition Accuracy for Pre-Boundary Frames

Prediction: NB participants would show higher accuracy for before-boundary (BB) frames than AB participants.

Rationale: The boundary advantage predicts that frames just before a natural boundary benefit from heightened perceptual and attentional processing. In the AB condition, the artificial early cut eliminates this encoding boost for BB frames.

Normality check. The Shapiro-Wilk test confirmed that BB-frame accuracy was significantly non-normal in both the AB group ($W = 0.946, p = .002$) and the NB group ($W = 0.925, p < .001$). Consequently, the independent-samples t-test was replaced by the **Mann-Whitney U test**.

Table 5.1. Mean BB-frame recognition accuracy

Group	M	SD	Mdn	n
AB	0.823	0.106	0.850	77
NB	0.855	0.096	0.850	87

NB participants ($M = 0.855$, $Mdn = 0.850$) achieved significantly higher BB-frame accuracy than AB participants ($M = 0.823$, $Mdn = 0.850$): $U = 2705.5$, $p = .032$, rank-biserial $r = 0.192$. This confirms that the natural encoding boost for boundary-adjacent frames is weakened when the boundary is cut short. A trial-level **Logistic GLM** (accuracy $\sim$ condition, BB trials only) corroborated this finding: being in the NB condition significantly increased the log-odds of a correct response ($B = 0.241$, $z = 2.526$, $p = .012$, $OR = 1.272$; $VIF_{\text{condition}} = 1.000$).

H2: Since $U = 2705.5$ and $p = .032 < \alpha = .05$ (GLM $p = .012$), we **reject the null hypothesis** and **support the alternative hypothesis** (H_A: NB participants show higher BB-frame accuracy than AB participants). Effect size is small-to-medium (rank-biserial $r = 0.192$).

5.3 H3 — Recognition Accuracy for Event-Middle Frames

Prediction: EM frame accuracy would not differ significantly between conditions.

Rationale: If the boundary effect only strengthens memory for moments immediately surrounding the boundary, frames from the event middle should be encoded similarly regardless of condition.

Normality check. The Shapiro-Wilk test confirmed non-normality for EM-frame accuracy in both the AB

group ($W = 0.923, p < .001$) and the NB group ($W = 0.914, p < .001$). Consequently, the **Mann-Whitney U test** was used.

Table 5.2. Mean EM-frame recognition accuracy

Group	M	SD	Mdn	n
AB	0.852	0.093	0.850	77
NB	0.884	0.080	0.900	87

Contrary to prediction, NB participants ($M = 0.884$, $Mdn = 0.900$) were also significantly more accurate on EM frames than AB participants ($M = 0.852$, $Mdn = 0.850$): $U = 2648.0$, $p = .019$, rank-biserial $r = 0.209$. A trial-level **Logistic GLM** (accuracy $\sim$ condition, EM trials only) confirmed the condition effect: being in the NB condition significantly increased accuracy log-odds ($B = 0.280$, $z = 2.701$, $p = .007$, $OR = 1.323$; $VIF_{\text{condition}} = 1.000$). The accuracy drop in the AB group was uniform across both BB and EM frames, suggesting that an abrupt cut impairs memory for the entire clip, not only boundary-adjacent content.

H3: Since $U = 2648.0$ and $p = .019 < \alpha = .05$ (GLM $p = .007$), we **fail to retain the null hypothesis** (H_0: no difference in EM-frame accuracy). Contrary to prediction, the boundary disruption generalises beyond BB frames.

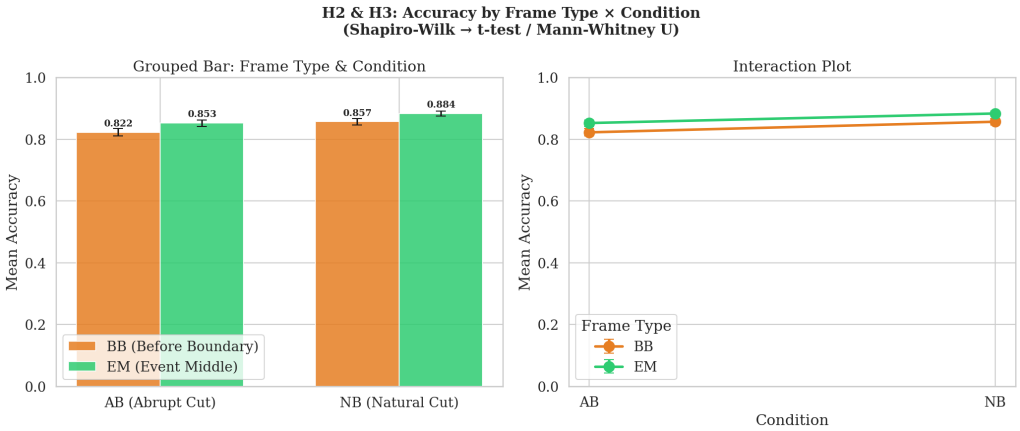

Figure 5.2 — H2 & H3: Mean recognition accuracy by frame type (BB, EM) and condition.

5.4 H4 — Confidence Calibration

Prediction: Within each condition (AB and NB), participants would report higher confidence on correct trials than on incorrect trials.

Rationale: If subjective confidence tracks genuine memory signal strength, it should be positively calibrated with accuracy. Analyzing this separately per condition confirms that metacognitive signals are robust regardless of boundary disruption.

One participant who made no errors across the session was excluded from this specific test.

We first evaluated the normality of the differences in confidence ratings using the Shapiro-Wilk test. For the AB group, the distribution was normal ($W = 0.975, p = .134$), leading to a paired t-test. For the NB group, the distribution was non-normal ($W = 0.969, p = .033$), requiring the Wilcoxon Signed-Rank test.

Table 5.3. H4 Results: Confidence Calibration by Condition

Group	Correct M	Incorr. M	Test	Stat	p	Effect Size
AB	4.201	3.362	t-test	$t = 12.562$	$< .001$	$d = 1.422$
NB	4.311	3.356	Wilcoxon	$W = 50.5$	$< .001$	$r = 0.971$

In both conditions, participants were significantly more confident when correct than when incorrect ($p <$

.001), with the effect consistently in the predicted direction (**Correct** > **Incorrect**). Effect sizes were very large ($d = 1.422, r = 0.971$).

H4: Since $p < .001$ for both groups and the direction is correct, we **reject the null hypothesis** and **strongly support the alternative hypothesis**.

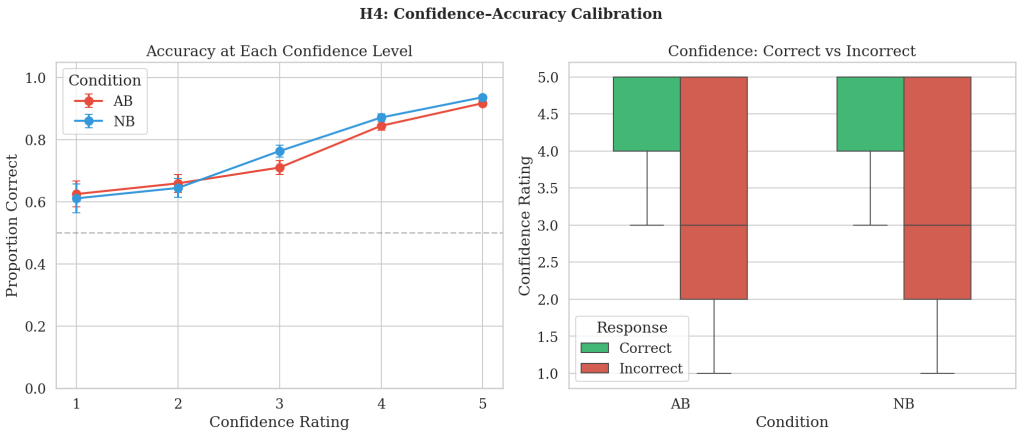

Figure 5.3 — H4: Mean confidence ratings for correct vs. incorrect trials by condition.

5.5 H5 — Confidence by Frame Type

Prediction: We expected that participants would feel more confident for BB frames than EM frames.

Data Cleaning: Removed trials where the RT was very high greater than $M + 4 \times SD$ as explained in 4.2.

Descriptive Statistics: Contrary, participants were not more confident for BB frames. In the AB condition confidence was lower for BB frames ($M = 4.03, SD = 0.52$) compared to EM frames ($M = 4.12, SD = 0.49$). In the NB condition, confidence was mostly same for BB frames ($M = 4.20, SD = 0.55$) and EM frames ($M = 4.22, SD = 0.47$).

Assumption Testing: Shapiro-Wilk test showed that the differences between BB and EM frames is a normal distribution for both groups (AB: $W = 0.978, p = .190$; NB: $W = 0.982, p = .265$). So we used paired t-tests to compare them.

Inferential Statistics: The paired t-test for the AB group showed a significant difference ($t(77) = -3.004, p = .0036$) but in the opposite direction than expected (EM > BB). For the NB group, the difference was not significant ($p = .4097$).

Trial-level OLS Regression supported the results showing that in the AB condition, confidence for BB frames was significantly lower ($B = -0.091, p = .025$). It also showed that overall confidence was higher in the NB group compared to the AB group ($B = 0.106, p = .008$).

Conclusion: Since confidence for BB frames was either lower or not different we reject both the null hypothesis (H_0) and the alternative hypothesis (H_A). Meaning natural event boundaries do not increase confidence and abrupt cuts actually reduce confidence for moments just before the cut.

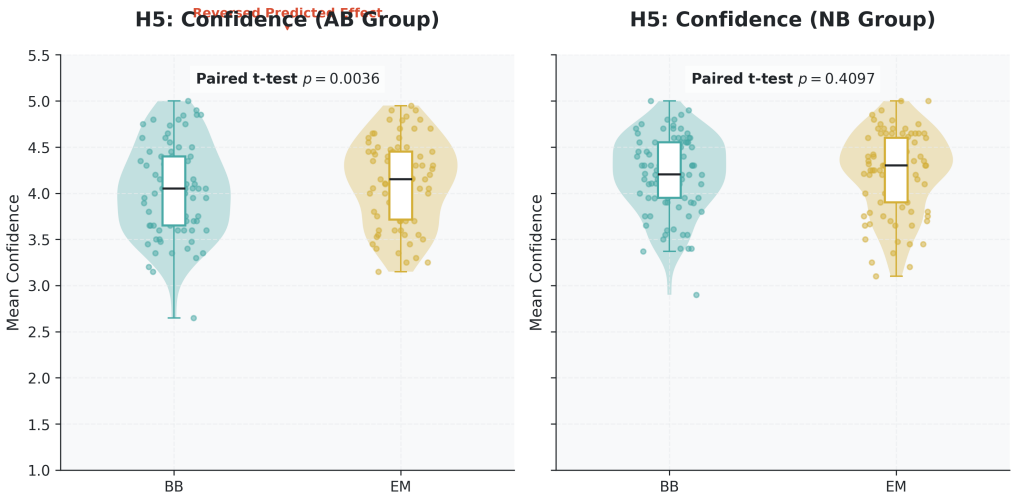

Figure 5.4 — H5: Mean confidence by frame type (BB vs. EM) and condition.

5.6 H6 — Demographic Equivalence

We first evaluated the normality of the age distribution using the Shapiro-Wilk test. Age was found to be significantly non-normal in both AB ($W = 0.855, p < .001$) and NB ($W = 0.929, p < .001$) groups. Consequently, an independent-samples Mann-Whitney U test was used for age. Categorical variables were tested using Chi-square tests of independence.

Table 5.5. Demographic Balance Checks

Variable	AB	NB	Test	Stat	p
Age	Mdn=22.3	Mdn=22.0	Mann-Whitney U	$U = 4391.5$	.564
Gender	51M, 27F	55M, 33F	Chi-square	$\chi^2(1) = 0.205$	.651
Handedness	71R, 7L	83R, 5L	Chi-square	$\chi^2(1) = 0.491$	.484
Vision	46N, 32C	45N, 43C	Chi-square	$\chi^2(2) = 2.161$	.339

There were no significant differences between groups on any demographic factor (all $p > .05$), confirming that random assignment produced balanced experimental groups.

H6: Since all $p > .05$, we **fail to reject the null hypothesis** and **support the alternative hypothesis** (groups are balanced).

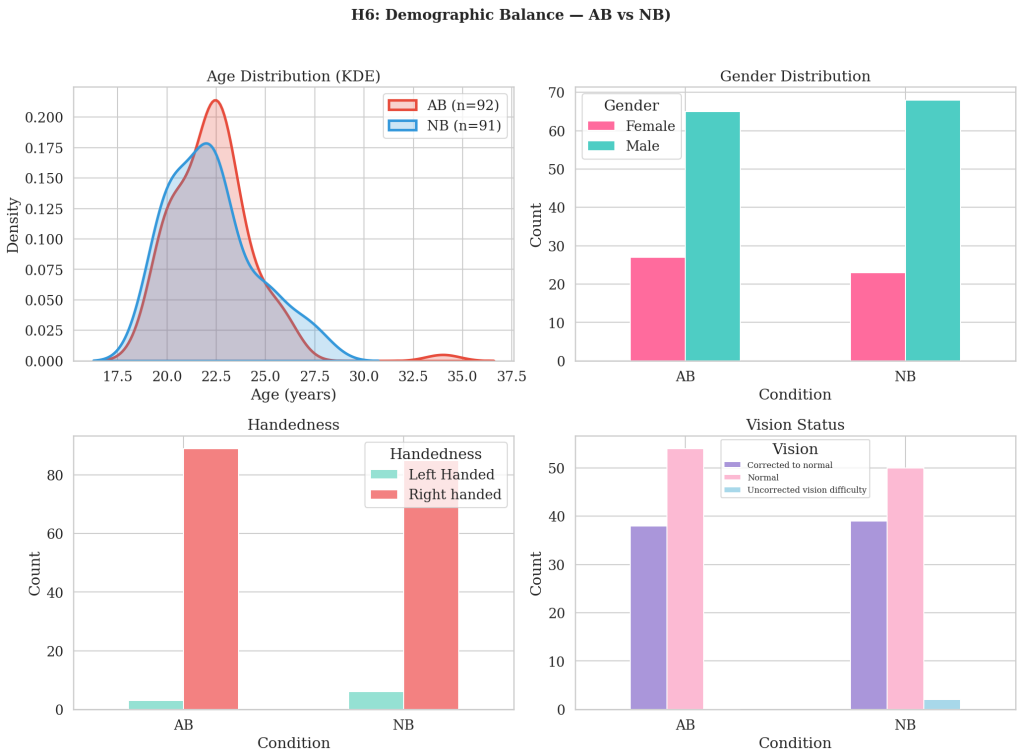

Figure 5.5 — H6: Age and gender distribution by condition.

Table 5.6. Participant-level means by condition

Variable	AB ($n = 78$)	NB ($n = 88$)
Accuracy (M, SD)	0.84 (0.08)	0.87 (0.07)
RT in s (M, SD)	5.57 (1.42)	5.75 (1.63)

6. Exploratory Analysis: Predicting Recognition Accuracy

While our previous tests looked at factors separately, we used a General Linear Model (Logistic Regression) to assess trials simultaneously ($N = 6,640$). This allowed us to test how confidence, experimental condition, and frame type (BB vs. EM) jointly predict accuracy.

Table 6.1. Logistic Regression Results for Recognition Accuracy

Predictor	Coef (B)	z	p	Odds Ratio
Intercept	-0.692	-5.99	$< .001$	0.50
Condition (NB)	0.183	2.51	.012	1.20
Frame Type (EM)	0.208	2.84	.004	1.23
Confidence	0.589	21.21	$< .001$	1.80

Confidence was the strongest independent predictor of accuracy; for every 1-point increase in subjective confidence, the odds of a correct answer increased by 80.2% ($OR = 1.80$). Both condition and frame type were also significant predictors, confirming that even when controlling for confidence, the natural-boundary advantage remains. Variance Inflation Factor (VIF) checks confirmed that all predictors were independent ($VIF \approx 1.0$), ensuring the model is stable.

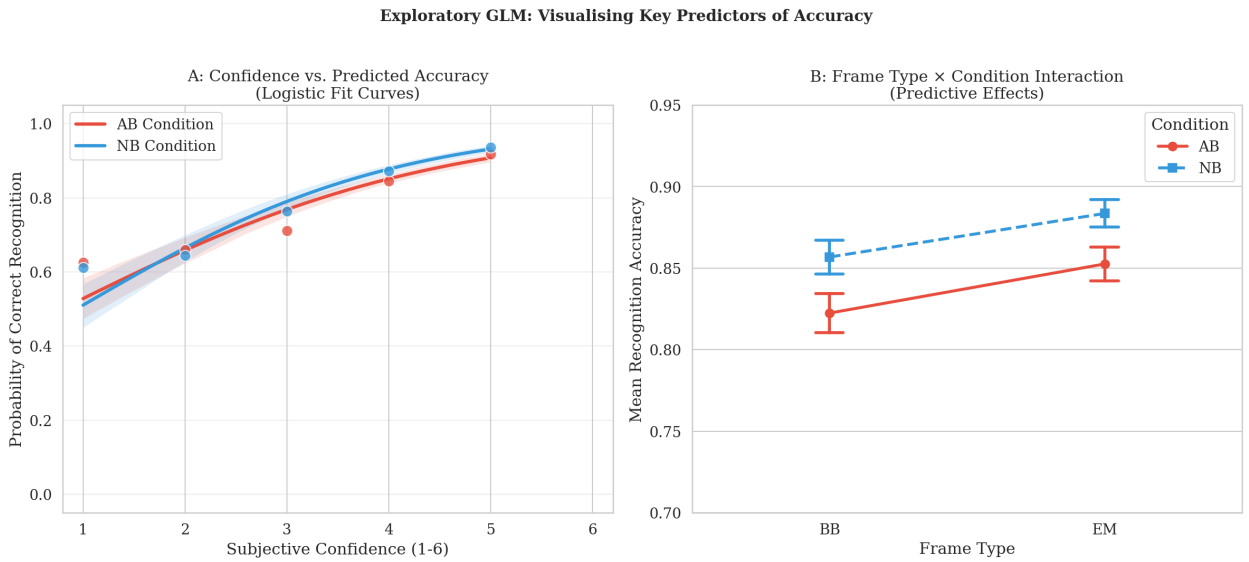

Figure 6.1: (A) Accuracy predicted by confidence; (B) Effect of video cuts and frame types.

7. Summary of Findings

- **Hypothesis 1 (Effect of Condition):** NB participants were significantly more accurate than AB participants ($U = 2550.5, p = .0042$), while no significant difference was found in response time ($p = .548$) confirming that preserving natural boundaries improves memory without speed-accuracy trade-off.
- **BB frames (H2):** Shapiro-Wilk confirmed non-normality in both groups; Mann-Whitney U ($U = 2705.5$, $p = .032$, rank-biserial $r = 0.192$) showed NB participants were significantly more accurate on BB frames. Trial-level GLM corroborated: $OR = 1.272$, $p = .012$. The boundary advantage for encoding-adjacent frames is real.
- **EM frames (H3):** Contrary to prediction, EM accuracy also differed significantly ($U = 2648.0$, $p = .019$, $r = 0.209$; GLM: $OR = 1.323$, $p = .007$). The boundary disruption impairs memory for the entire clip, not only boundary-adjacent frames.
- **Confidence (H4):** Strongest result — participants were substantially more confident when correct than incorrect (AB: $d = 1.42$; NB: $r = 0.97$), demonstrating reliable metacognitive sensitivity.
- **Confidence by frame type (H5):** The AB group showed *lower* confidence for BB than EM frames — the opposite of the prediction. No effect in the NB group.
- **Demographics (H6):** Both groups were perfectly matched on all demographic factors (age, gender, handedness, and vision). No sample imbalances were found.
- **Exploratory Analysis (GLM):** Using a multi-factor model with $VIF \approx 1.0$, we found that confidence, condition, and frame type are all independent, significant drivers of recognition accuracy. Confidence is the most powerful predictor, yielding an 80.2% increase in the odds of accuracy per point.

8 References

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9. Contributions

- **Akash** — Data cleaning, Analysis for H1 and H5, report writing.
- **Varun** — Analysis for H4 and H6, Exploratory Analysis, report writing
- **Nikhilesh** — Analysis for H2 and H3, report writing.

Code Repository. https://github.com/Varun2345/BRSM-Project